Independent Model Validation Report: PD Model (Champion Logistic) with Challenger Benchmark (LightGBM)

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Executive Summary

- **Scope:** Independent validation of a probability-of-default (PD) model using LendingClub loan data, covering performance, calibration, out-of-time (OOT) generalization, drift monitoring (PSI/KS), sensitivity and stress testing, and a challenger benchmark.
- **Champion (logistic):** The champion shows reasonable OOT discrimination, but calibration degrades OOT, with predicted PD levels above the observed default rate, consistent with population and base-rate shift.
- **Drift:** PSI highlights material drift in key underwriting variables, especially revolving utilization (`revol_util`), with additional drift in `application_type` and `int_rate`.
- **Robustness:** Sensitivity analysis shows highly stable ranking under small shocks (Spearman ≈ 1), while `dti` is the most sensitivity-relevant driver among the tested variables.
- **Challenger (LightGBM):** The challenger improves OOT discrimination and calibration characteristics and shows lower score drift, supporting promotion as a redevelopment candidate subject to governance controls.

1 Scope and Model Purpose

1.1 Model Purpose

The validated model estimates a **Probability of Default (PD)** for unsecured consumer loans. PD is intended to support risk segmentation, portfolio monitoring, and (with appropriate calibration governance) downstream loss estimation and decision support.

1.2 Validation Scope

This validation covers the following components:

- **Reproducibility:** Replication of the "bank" champion specification (logistic regression) under a controlled feature policy.
- **Performance:** Discrimination metrics (AUC, KS, PR-AUC, top-decile capture) on Train and OOT.
- **Calibration:** Reliability curves and calibration-in-the-large / calibration slope diagnostics.
- **OOT validation:** Explicit time-based split with Train vs OOT comparison.

- **Drift monitoring:** PSI for inputs and score; two-sample KS on score distributions.
- **Sensitivity:** $\pm 5\%$ and $\pm 10\%$ shocks on key drivers, including ranking stability via Spearman correlation.
- **Stress testing:** Mild and severe macro-style overlays, with Expected Loss proxy ($PD \times LGD \times EAD$).
- **Challenger benchmark:** LightGBM challenger comparison plus interpretability (SHAP summary bar plot).

2 Data, Target Definition, and Feature Governance

2.1 Dataset

This report uses the LendingClub public loan dataset (Kaggle). The dataset contains borrower/application variables, origination dates, and loan outcomes (e.g., `loan_status`), as well as some post-origination fields (e.g., recoveries, payment variables).

2.2 Time Split (OOT Design)

A true **out-of-time (OOT)** validation is implemented using `issue_d` (loan origination date). In the current dataset version used for this project:

- **Train:** historical vintages up to 2017 (inclusive)
- **OOT:** 2018 vintage

This split is designed to simulate realistic deployment conditions where the model is trained on past vintages and evaluated on a later population.

2.3 PD Target Definition

To avoid label contamination from unresolved loans, the PD target is defined only on **closed outcomes**:

- `default_flag = 1`: `Charged Off` or `Default`
- `default_flag = 0`: `Fully Paid`
- Excluded: `Current`, `Late`, `In Grace Period`, and other statuses with incomplete final outcome information

This is a standard validation choice when the goal is observed default discrimination/calibration on resolved loans.

2.4 Feature Policy and Leakage Prevention

Inputs are restricted to **underwriting-time variables** only. Post-outcome variables (payments, recoveries, outstanding balances, settlement-related columns, etc.) are excluded to prevent **data leakage**.

Leakage control is implemented through:

- **Whitelist-based feature policy** for model inputs (pre-origination / underwriting variables only)
- **Automated diagnostic scan** for leakage candidate column names (e.g., payment, recovery, settlement, hardship patterns)

This governance design is important because strong performance driven by post-outcome variables would invalidate the model risk assessment.

3 Champion Model Reproduction (Logistic Regression)

3.1 Champion Specification

The champion model is implemented as a **logistic regression pipeline** with explicit preprocessing:

- **Numeric features:** Train-based median imputation + standardization (improves optimizer convergence and coefficient stability)
- **Categorical features:** Imputation + one-hot encoding with unknown-category handling
- **Missingness governance:** Features with excessive missingness in Train are excluded based on a documented threshold policy

This setup replicates a realistic "bank champion" baseline while remaining auditable and reproducible.

3.2 Performance and Calibration Summary (Train vs OOT)

Table 1 summarizes performance and calibration metrics for both the Champion and the Challenger across Train and OOT. The table is included here to anchor the report's quantitative conclusions.

Table 1: Performance and calibration summary (Train vs OOT).

Part 1/3

model	set	AUC	PR_AUC	Gini
Champion_Logit	Train	0.710400	0.382400	0.420800
Champion_Logit	OOT	0.697300	0.263100	0.394700
Challenger_LGBM	Train	0.731800	0.415500	0.463500
Challenger_LGBM	OOT	0.726200	0.303800	0.452500

Part 2/3

model	KS	Brier	Top10pct_Default_Capture	Mean_PD
Champion_Logit	0.304800	0.148500	0.225200	0.206700
Champion_Logit	0.293900	0.121300	0.209200	0.190600
Challenger_LGBM	0.336800	0.144800	0.242100	0.206700
Challenger_LGBM	0.337200	0.118300	0.241100	0.196800

Part 3/3

model	Obs_Default_Rate	Calib_Intercept	Calib_Slope
Champion_Logit	0.206700	-0.000500	0.999600
Champion_Logit	0.147300	-0.542000	0.837000
Challenger_LGBM	0.206700	0.091500	1.079000
Challenger_LGBM	0.147300	-0.418500	0.973900

Validation interpretation. The champion maintains reasonable discrimination OOT, but calibration deteriorates (predicted mean PD above observed default rate, and weaker calibration

slope/intercept). This pattern is consistent with a shift in population characteristics and/or outcome prevalence between Train and OOT. At the same time, the summary table indicates that the challenger preserves stronger OOT performance and calibration characteristics, which motivates the dedicated challenger assessment in Section 6.

3.3 Champion Calibration Evidence

Reliability plots compare binned predicted PDs against observed default rates.

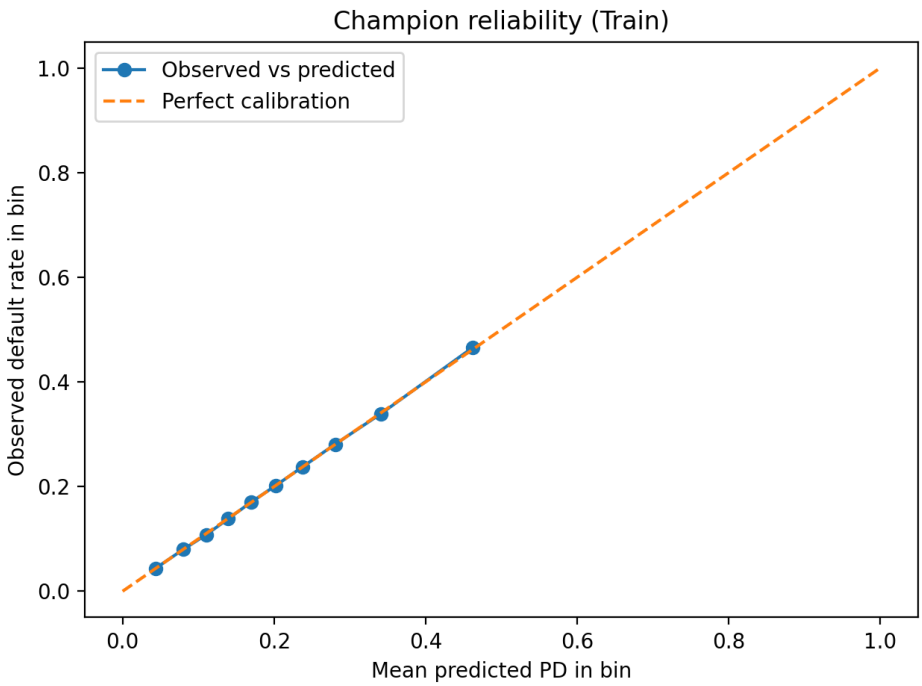

Figure 1: Champion reliability curve (Train).

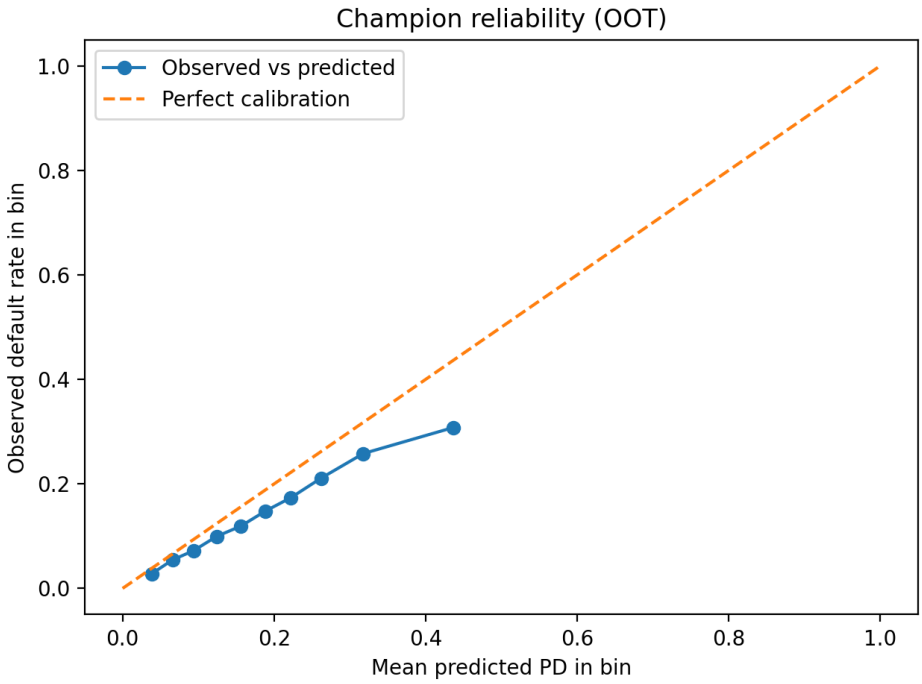

Figure 2: Champion reliability curve (OOT).

Validation interpretation. Train calibration is materially stronger than OOT calibration. In OOT, the reliability curve sits below the perfect-calibration line across most bins, indicating that predicted PDs are generally above observed default rates (i.e., conservative/over-predictive calibration). This supports a standard validator conclusion: the model continues to rank risk reasonably well, but probability calibration should be monitored and periodically recalibrated as portfolio conditions evolve.

4 OOT Validation and Drift Monitoring

4.1 Score Distribution Shift

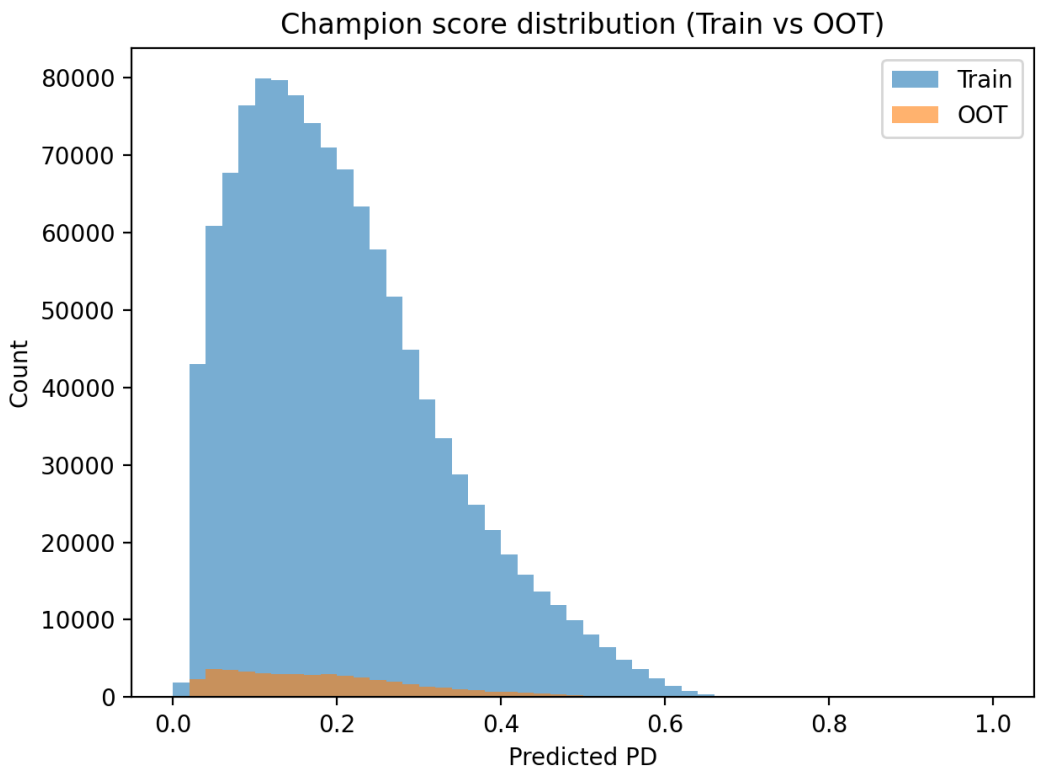

Figure 3: Champion score distribution shift (Train vs OOT).

The score distribution changes between Train and OOT, which is expected when portfolio composition and underwriting or macro conditions shift over time. Distribution shift does not automatically imply model failure, but it increases the importance of calibration and drift monitoring.

4.2 Population Stability Index (PSI)

PSI is used as the primary drift indicator for features and score distributions. Standard thresholds are applied:

- PSI < 0.10: stable (GREEN)
- $0.10 \leq$ PSI < 0.25: moderate drift (AMBER)
- PSI ≥ 0.25: material drift (RED)

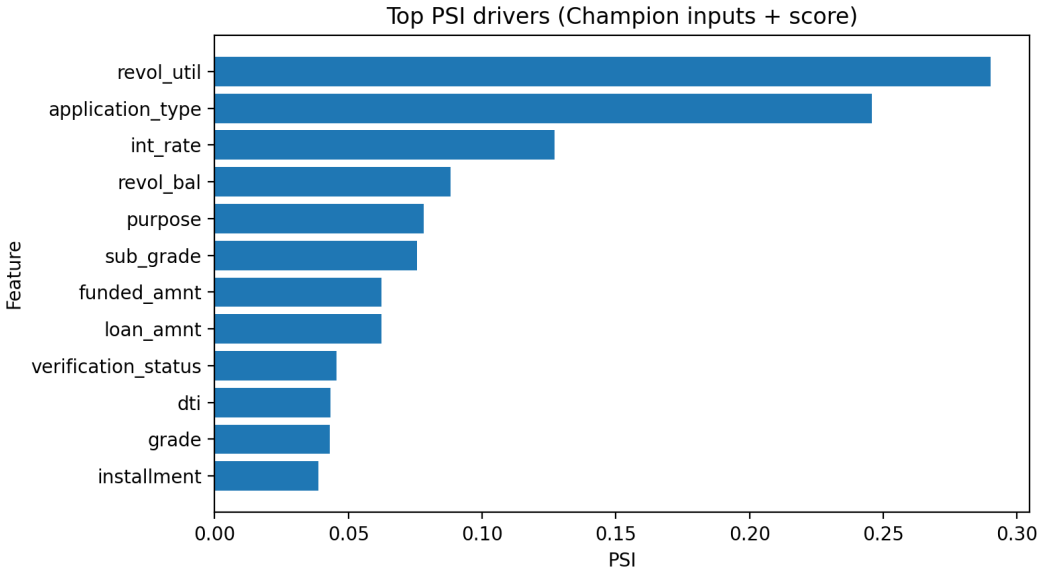

Figure 4: Top PSI drift drivers (Champion inputs + score).

Table 2: Top PSI drift drivers (Champion inputs and score).

feature	type	psi	flag
revol_util	numeric	0.290200	RED
application_type	categorical	0.245700	AMBER
int_rate	numeric	0.127100	AMBER
revol_bal	numeric	0.088100	GREEN
purpose	categorical	0.078300	GREEN
sub_grade	categorical	0.075600	GREEN
funded_amnt	numeric	0.062500	GREEN
loan_amnt	numeric	0.062500	GREEN
verification_status	categorical	0.045600	GREEN
dti	numeric	0.043200	GREEN
grade	categorical	0.043200	GREEN
installment	numeric	0.038700	GREEN
pub_rec	numeric	0.037600	GREEN
pd_pred	score	0.025900	GREEN
emp_length	categorical	0.021800	GREEN
total_acc	numeric	0.020400	GREEN
open_acc	numeric	0.019000	GREEN
credit_history_length_years	numeric	0.014500	GREEN
delinq_2yrs	numeric	0.010600	GREEN
inq_last_6mths	numeric	0.009800	GREEN
home_ownership	categorical	0.009600	GREEN
annual_inc	numeric	0.005900	GREEN
term	categorical	0.004600	GREEN

Key drift finding. The strongest drift is observed in `revol_util` (material/RED), with additional drift in `application_type` and `int_rate`. These are plausible drift drivers from a credit-risk perspective (portfolio mix, pricing, utilization behavior, credit-cycle effects). The evidence supports ongoing drift monitoring and reinforces the calibration findings.

4.3 KS Score Shift (Train vs OOT)

In addition to PSI, a two-sample KS test is used on the champion score distribution to quantify Train–OOT score shift. The result indicates a visible but not extreme distribution change; given the large sample size, the p-value is less decision-relevant than the KS effect size itself. This diagnostic is used as supporting evidence alongside score PSI and OOT calibration.

5 Sensitivity and Stress Testing

5.1 Sensitivity Analysis

Sensitivity tests perturb key numeric drivers by ±5% and ±10% and assess:

- change in mean predicted PD
- change in PD tail (95th percentile)
- rank-order stability (Spearman correlation)

Table 3: Sensitivity analysis (OOT): PD shift and ranking stability under input shocks.

Part 1/2

feature	shock	mean_pd_base	mean_pd_new	delta_mean_pd
annual_inc	-0.100000	0.190617	0.192223	0.001606
annual_inc	-0.050000	0.190617	0.191418	0.000801
annual_inc	0.050000	0.190617	0.189820	-0.000797
annual_inc	0.100000	0.190617	0.189027	-0.001591
dti	-0.100000	0.190617	0.186945	-0.003673
dti	-0.050000	0.190617	0.188775	-0.001842
dti	0.050000	0.190617	0.192470	0.001853
dti	0.100000	0.190617	0.194335	0.003717
revol_util	-0.100000	0.190617	0.190006	-0.000611
revol_util	-0.050000	0.190617	0.190311	-0.000306
revol_util	0.050000	0.190617	0.190915	0.000298
revol_util	0.100000	0.190617	0.191202	0.000585

Part 2/2

feature	p95_base	p95_new	delta_p95	spearman_rank_corr
annual_inc	0.417995	0.420332	0.002337	0.999958
annual_inc	0.417995	0.419096	0.001101	0.999989
annual_inc	0.417995	0.416774	-0.001222	0.999990
annual_inc	0.417995	0.415630	-0.002365	0.999959
dti	0.417995	0.410241	-0.007754	0.999867
dti	0.417995	0.414109	-0.003886	0.999967
dti	0.417995	0.422023	0.004028	0.999968
dti	0.417995	0.425906	0.007911	0.999872
revol_util	0.417995	0.416765	-0.001231	0.999995
revol_util	0.417995	0.417401	-0.000595	0.999999
revol_util	0.417995	0.418719	0.000724	0.999999
revol_util	0.417995	0.419449	0.001454	0.999996

Key sensitivity finding. Ranking remains highly stable under all tested shocks (Spearman approximately 1), indicating robust rank ordering. Among the tested variables, `dti` shows the largest impact on mean PD and tail PD, making it the most sensitivity-relevant driver in this analysis.

5.2 Stress Scenarios and Expected Loss Proxy

Two stylized scenarios are applied to OOT observations:

- **Mild recession:** lower income and higher DTI
- **Severe recession:** larger income reduction, larger DTI increase, and higher utilization (capped)

An Expected Loss proxy is computed as:

$$EL_{\text{proxy}} = PD_{\text{pred}} \times LGD_{\text{avg}} \times EAD_{\text{proxy}}$$

with:

- $EAD_{\text{proxy}} \approx \texttt{funded_amnt} - \texttt{total_rec_prncp}$ (if available), otherwise `funded_amnt`
- $LGD_{\text{proxy}} \approx \max\{0, \min(1, 1 - \texttt{recoveries}/EAD_{\text{proxy}})\}$ for defaulted loans
- LGD_{avg} as the empirical average of LGD_{proxy} over defaults (proxy-based, not a calibrated LGD model)

Table 4: Stress scenarios (OOT): PD and EL proxy shifts.

Part 1/2

scenario	mean_pd	delta_mean_pd	p95_pd	delta_p95
mild	0.195145	0.004528	0.427175	0.009180
severe	0.200342	0.009725	0.436968	0.018973

Part 2/2

scenario	EL_proxy_mean	delta_EL_proxy	LGD_avg_defaults
mild	647.706794	13.718138	0.979338
severe	663.375656	29.387000	0.979338

Key stress finding. Both mean PD and tail PD increase monotonically from baseline to mild to severe stress, and the EL proxy rises accordingly. This is the expected qualitative behavior and supports the model's use in vulnerability analysis (with documented proxy limitations).

5.3 LGD Proxy Profile (Defaults Only)

Table 5: LGD proxy by grade (defaults only): summary statistics.

grade	count	mean	median	p90
C	2128	0.980893	1.000000	1.000000
D	1974	0.977382	1.000000	1.000000
B	1402	0.981952	1.000000	1.000000
E	707	0.974824	1.000000	1.000000
A	519	0.983331	1.000000	1.000000
F	175	0.971131	1.000000	1.000000
G	46	0.967148	1.000000	1.000000

Interpretation. LGD proxy levels are high in this dataset, consistent with low recoveries in many charged-off unsecured consumer loans. This is useful for *directional* stress and EL proxy analysis, but it should not be interpreted as a production-ready regulatory LGD model.

6 Challenger Benchmark (LightGBM)

6.1 Challenger Rationale

A LightGBM challenger is introduced to test whether a non-linear model improves:

- OOT discrimination
- calibration characteristics
- score stability (lower score drift)

The challenger is trained using the same governance-approved feature set as the champion to preserve comparability and avoid unfair performance gains from different information sets.

6.2 OOT Champion vs Challenger Comparison

Table 6: Out-of-time comparison: Champion vs Challenger.

Part 1/3

model	AUC	PR_AUC	KS	Brier
Champion_Logit	0.697300	0.263100	0.293900	0.121300
Challenger_LGBM	0.726200	0.303800	0.337200	0.118300

Part 2/3

model	Top10pct_Default_Capture	Mean_PD	Obs_Default_Rate	Calib_Intercept
Champion_Logit	0.209200	0.190600	0.147300	-0.542000
Challenger_LGBM	0.241100	0.196800	0.147300	-0.418500

Part 3/3

model	Calib_Slope
Champion_Logit	0.837000
Challenger_LGBM	0.973900

Decision-relevant finding. The challenger improves OOT discrimination (AUC / PR-AUC) and shows better calibration characteristics (slope closer to 1, less adverse calibration intercept) than the champion. This suggests an improvement that is not purely in-sample overfitting.

6.3 Score Stability Comparison

Table 7: Score stability summary (Train to OOT).

model	score_psi	score_ks_stat	score_ks_p
Champion_Logit	0.025877	0.057605	0.000000
Challenger_LGBM	0.015908	0.044408	0.000000

Stability finding. The challenger exhibits lower score drift (lower score PSI and lower KS score shift), which strengthens the redevelopment case from a model risk perspective.

6.4 Interpretability (SHAP)

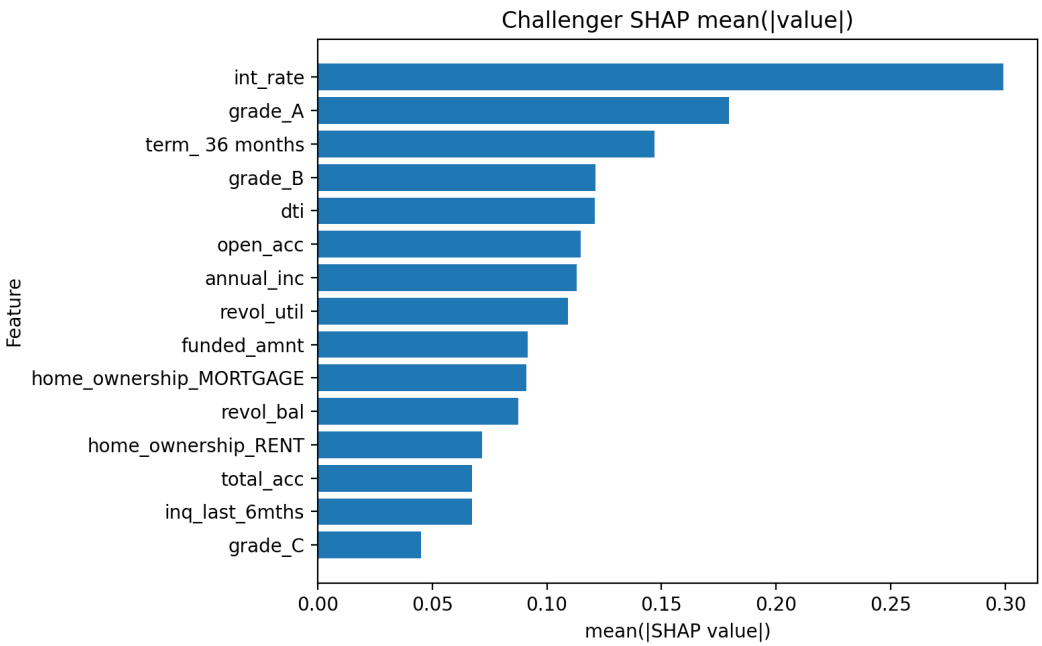

Figure 5: Challenger interpretability (SHAP mean absolute contribution).

Sanity check. Top challenger drivers align with credit intuition (e.g., rate, utilization, DTI, grade/term-related effects). No post-origination leakage-type variables appear among the major drivers, which supports feature governance compliance.

7 Conclusion, Model Risk Rating, and Action Plan

7.1 Proposed Model Risk Rating

Proposed rating: Moderate Model Risk (Champion).

Rationale.

- The champion demonstrates **reasonable OOT discrimination**, so ranking value remains present.
- The champion shows **OOT calibration degradation**, with PD levels overpredicting the observed OOT default rate and weaker calibration slope/intercept.
- **Drift is material** in key inputs (notably `revol_util`), consistent with a real population shift.
- Sensitivity results show **strong ranking robustness** under small perturbations, which mitigates operational fragility concerns.

This combination supports a **Moderate** rating rather than Low (calibration + drift issues present) or High (discrimination remains acceptable and behavior under stress/sensitivity is coherent).

7.2 Recommended Remediation and Monitoring Plan

1. **Drift monitoring (monthly/quarterly):**
   Track PSI for key drivers (`revol_util`, `application_type`, `int_rate`) and score (`pd_pred`); investigate AMBER/RED threshold breaches.
2. **Calibration governance:**
   Monitor calibration-in-the-large and slope on recent vintages; apply periodic recalibration if predicted PD levels materially diverge from observed default rates.
3. **Segment monitoring:**
   Monitor discrimination and calibration by key segments (e.g., loan term, high-volume purpose categories, grade groups), especially where OOT degradation is concentrated.
4. **Challenger governance (if promoted):**
   Apply the same validation suite (performance, calibration, drift, sensitivity, stress) to the challenger in production monitoring, and maintain periodic interpretability review (SHAP/global importance) as part of governance documentation.
5. **Documentation and limitations:**
   Explicitly document the proxy nature of EAD/LGD/EL calculations, stress assumptions, and the fact that the LGD/EAD module is intended for directional model-risk analysis rather than regulatory capital modeling.

Ownership Signals Delivered

- Reproduced the champion ("bank") model end-to-end with controlled preprocessing and leakage governance.
- Performed independent OOT validation with explicit Train/OOT temporal split.
- Assessed discrimination, calibration, drift (PSI/KS), sensitivity, and stress behavior.
- Benchmarked a challenger model with interpretability and stability evidence.

- Extended the validation pack to PD–LGD–EAD proxy-based EL analysis to demonstrate full credit risk stack awareness.